

IRSSG Installation and User Manual

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1 Program Overview

IRSSG is an open-source software package designed for magnetic crystals with spin space groups (SSGs) in the weak spin-orbit-coupling regime. Starting from either *ab initio* wavefunctions (for example VASP WAVECAR files) or Wannier tight-binding Hamiltonians, the program automatically identifies all SSG operations, SSG numbers, and international symbols for a given magnetic structure, constructs SSG k -little-group character tables at high-symmetry k points, and computes traces of SSG matrix representations acting on the bands so as to assign irreducible corepresentation labels to each magnetic band. With a suitable `ssg.data` file, IRSSG can also be applied to ordinary crystallographic and magnetic space groups, and is useful for analyzing high-degeneracy points, magnon and electronic band topology, altermagnets, and related magnetic-topological phases.

2 Installation

2.1 Python Requirements

IRSSG is distributed as a Python package and requires:

- Python 3.8–3.13;
- `numpy` $\geq$ 1.20.1;
- `scipy` $\geq$ 1.5.4;
- `spglib` $\geq$ 1.15.0;
- `pymatgen` $\geq$ 2021.2.8;
- `phonopy` $\geq$ 2.7.0;

2.2 Installing via pip

With internet access, installation can be completed directly using `pip`:

```
$ pip install irssg
```

When executing the above command, `pip` will automatically attempt to install all required dependencies. After installation, the command `irssg` should be directly available in the command line, which typically indicates successful installation.

If errors related to missing dependencies or compilation occur during `pip install irssg`, it is recommended to manually install the dependencies first, then re-run `pip install irssg`. In most environments, one can use `conda` to install common numerical libraries first, for example:

```
$ conda install numpy h5py contourpy pandas  
$ pip install irssg
```

In general, the above combination can resolve the vast majority of dependency-related issues.

3 Command-Line Usage and Parameter Descriptions

3.1 `irssg -ssg`: SSG Analysis of Materials

The `-ssg` mode can be used when one only needs to identify spin space group operations, generate SSG numbers, and SSG international symbols.

3.1.1 Input Files

The `-ssg` mode requires an input file POSCAR or `*.mcif`. For each magnetic atom in POSCAR, the magnetic moment (m_x, m_y, m_z) in Cartesian coordinates should be written after the atomic coordinates, with spaces between all entries, so the corresponding line contains 6 numbers in total. If the magnetic moment on an atom is zero, the line may contain only the 3 coordinate values and omit the magnetic moment.

A typical example is Mn_3Sn :

```
Mn3Sn
1.0
5.665000 0.000000 0.000000
-2.832500 4.906034 0.000000
0.000000 0.000000 4.531000
Mn Sn
6 2
Direct
0.838800 0.677600 0.250000 1.5 2.5981 0
0.161200 0.838800 0.750000 -3 0 0
0.838800 0.161200 0.250000 -3 0 0
0.161200 0.322400 0.750000 1.5 2.5981 0
0.322400 0.161200 0.250000 1.5 -2.5981 0
0.677600 0.838800 0.750000 1.5 -2.5981 0
0.333333 0.666667 0.250000
0.666667 0.333333 0.750000
```

Another typical example is the altermagnetic material $\text{KV}_2\text{Se}_2\text{O}$:

```
KV2Se2O
1.0
3.951940 0.000000 0.000000
0.000000 3.951940 0.000000
0.000000 0.000000 7.352430
K V Se O
1 2 2 1
Direct
0.000000 0.000000 0.500000
0.500000 0.000000 0.000000 0 0 3
0.000000 0.500000 0.000000 0 0 -3
0.500000 0.500000 0.240450
0.500000 0.500000 0.759550
0.000000 0.000000 0.000000
```

3.1.2 Command

```
$ irssg -ssg [SSG-related options]
```

By default, the program reads the crystal structure with spins from POSCAR (or *.mcif) in the current directory and performs SSG analysis. Common options include:

- **-c FILE:** Specify the input structure file name (default POSCAR; *.mcif files are also supported);
- **--tolerance TOL:** Tolerance for spatial symmetry operations (default 10^{-3});
- **--magtolerance MTOL:** Tolerance for spin/magnetic moments (default 10^{-4});
- **--standardize:** Standardize the structure after analysis;

Typical example:

```
$ irssg -ssg -c POSCAR > ssg.out
```

3.1.3 Output Files

ssg.out (screen output)

This output file sequentially lists the file type (POSCAR or *.mcif), SSG number, SSG international symbol, spin-only group $\mathcal{S}_0$, Schönflies notation of the spin-part group $\mathcal{P}$, Hermann–Mauguin notation of the lattice-part group $\mathcal{H}$, SSG operations ($\mathcal{G}_0$), and other information. SSG operations ($\mathcal{G}_0$) are denoted as $\mathcal{O} = \{U||R|v\}$.

For type II (coplanar magnetic moments), the spin part U of the SSG is represented by a 2×2 O(2) matrix ζ , $U = \zeta \oplus I_1$; for type I (collinear magnetic moments), the spin part U of the SSG is represented by $\xi = 1$ (magnetic moment unchanged) or $\xi = -1$ (magnetic moment flipped), $U = I_2 \oplus \xi$.

A typical example for Mn_3Sn :

```
Spin space group (SSG) information is output below:
The SSG number: 194.1.6.1.P
The SSG international symbol: P~{1,1,1} 6_3~{3~-1} / m~1 m~m c~m (C3v~II)
II: Coplanar SSG in the x'y' plane; So: Cs={E,Mz'}
The redefined axes in spin space: (x',y',z')=(x,y,z)D
D = -0.000 -1.000 0.000
    1.000 -0.000 0.000
    0.000 0.000 1.000
The SSG G = So x Go
P (spin part of Go): C3v
H (lattice part of Go): P6_3/mmc
=====
Spin space group operations: {U||R|v}
U is given in x'y'z' coordinates, while R is given in the lattice basis of POSCAR.
U=  $\zeta_{2x2} + I_1$  in type II
# Number: 24
{  $\zeta_{2x2}$  || Ri | taui }
# 1 without time-reversal
1.000 0.000 1 0 0 0.000
0.000 1.000 0 1 0 0.000
0 0 1 0.000
.....
# 24 with time-reversal
0.500 -0.866 -1 1 0 -0.000
-0.866 -0.500 0 1 0 0.000
0 0 1 0.000

D U D~{-1} is in Cartesian coordinates, which is in ssg.data.
Atomic space group: 194, P6_3/mmc
```

```

Atomic primitive cell
a: [ 5.665, 0.000, 0.000]^T
b: [-2.832, 4.906, 0.000]^T
c: [ 0.000, 0.000, 4.531]^T

H:(a1,b1,c1) = (a,b,c)R1, det(R1) = 1   T0:(a2,b2,c2) = (a,b,c)R2, det(R2) = 1   POSCAR:(a3,b3,c3) = (a,b,c)R3, det(R3) = 1
R1 = 1.000 0.000 0.000      R2 = 1.000 0.000 0.000      R3 = 1.000 0.000 0.000
      0.000 1.000 0.000      0.000 1.000 0.000      0.000 1.000 0.000
      0.000 0.000 1.000      0.000 0.000 1.000      0.000 0.000 1.000

N_ASQ/N_SSG = 1
=====
More information about this SSG can be found at
https://cmpdc.iphy.ac.cn/hsp/
https://cmpdc.iphy.ac.cn/ssg/ssgs/194.1.6.1.P
=====
Magnetic space group (MSG) information is output below:
The MSG number: 63.7.517 (OG setting), 63.463 (BNS setting), 63.1.4.61 (SSG setting)
The MSG international symbol: Cmc'm'
Magnetic space group operations: {U||R|v}
{  U      ||  Ri  | tau_i }
# Number: 8
# 1 without time-reversal
1.000 0.000 0.000 1 0 0 0.000
0.000 1.000 0.000 0 1 0 0.000
0.000 0.000 1.000 0 0 1 0.000
.....
# 8 with time-reversal
-0.500 -0.866 0.000 -1 0 0 0.000
-0.866 0.500 0.000 -1 1 0 0.000
0.000 0.000 1.000 0 0 1 -0.500

Indices of MSG operations within the list of SSG operations:
1      2      7      8      15     16     21     22

More information about this MSG can be found at
https://cmpdc.iphy.ac.cn/hsp/
https://www.cryst.ehu.es

```

From this output, Mn_3Sn has SSG number 194.1.6.1.P. Its international symbol is $P^{1,1,1}6_3^{-1}/m^1m^m c^m(C_{3v}^{II})$. In addition, the point group $\mathcal{P}$ formed by the spin part of its SSG is C_{3v} , the space group formed by the spatial part is $P6_3/\text{mmc}$, and the spin-only group is C_s , indicating a coplanar magnetic configuration.

An example output for $\text{KV}_2\text{Se}_2\text{O}$ is as follows:

```

Spin space group (SSG) information is output below:
The SSG number: 123.1.2.7.L
The SSG international symbol:  $P^{1,1,1}4_2^m / m^1 m^1 m^m (\text{Cs}^1\text{I})$ 
I: Collinear SSG along z' direction; So:  $\omega\text{Cz}', \text{Mx}'\omega\text{Cz}'$ 
The redefined axes in spin space: (x',y',z')=(x,y,z)D
D = 0.000 -1.000 0.000
      1.000 0.000 0.000
      0.000 0.000 1.000
The SSG G = So x Go
P (spin part of Go): Cs
H (lattice part of Go): P4/mmm
=====
Spin space group operations: {U||R|v}
U is given in x'y'z' coordinates, while R is given in the lattice basis of POSCAR.
U= I_2 +  $\xi$  in type I
# Number: 16
{  $\xi$  || Ri | tau_i }
# 1 without time-reversal
1 1 0 0 0.000
0 1 0 0.000
0 0 1 0.000
.....
# 16 with time-reversal
-1 0 -1 0 0.000
-1 0 0 0.000
0 0 1 0.000

```

```

D U D^{-1} is in Cartesian coordinates, which is in ssg.data.

Atomic space group: 123, P4/mmm
Atomic primitive cell
a: [ 3.952, 0.000, 0.000]^T
b: [ 0.000, 3.952, 0.000]^T
c: [ 0.000, 0.000, 7.352]^T

H:(a1,b1,c1) = (a,b,c)R1, det(R1) = 1   T0:(a2,b2,c2) = (a,b,c)R2, det(R2) = 1   POSCAR:(a3,b3,c3) = (a,b,c)R3, det(R3) =
1
R1 = 1.000 0.000 0.000           R2 = 1.000 0.000 0.000           R3 = 1.000 0.000 0.000
      0.000 1.000 0.000           0.000 1.000 0.000           0.000 1.000 0.000
      0.000 0.000 1.000           0.000 0.000 1.000           0.000 0.000 1.000

N_ASQ/N_SSG = 1
=====
More information about this SSG can be found at
https://cmpdc.iphy.ac.cn/hsp/
https://cmpdc.iphy.ac.cn/ssg/ssgs/123.1.2.7.L
=====
Magnetic space group (MSG) information is output below:
The MSG number: 123.4.1002 (OG setting), 123.342 (BNS setting)
The MSG international symbol: P4'/mm'm
Magnetic space group operations: {U||R|v}
{   U           ||   Ri   |   tau_i }
# Number: 16
# 1 without time-reversal
1.000 0.000 0.000 1 0 0 0.000
0.000 1.000 0.000 0 1 0 0.000
0.000 0.000 1.000 0 0 1 0.000
.....
# 16 without time-reversal
0.000 1.000 0.000 0 -1 0 0.000
1.000 0.000 0.000 -1 0 0 0.000
0.000 0.000 -1.000 0 0 1 0.000

Indices of MSG operations within the list of SSG operations:
      1      2      3      4      5      6      7      8
      9     10     11     12     13     14     15     16

More information about this MSG can be found at
https://cmpdc.iphy.ac.cn/hsp/
https://www.cryst.ehu.es

```

From this output, $\text{KV}_2\text{Se}_2\text{O}$ has SSG number 123.1.2.7.L. Its international symbol is $P^{1,1,1}4^m/m^1m^1m^m(C_s^I)$. In addition, the point group $\mathcal{P}$ formed by the spin part of its SSG is C_s , the space group formed by the spatial part is $P4/mmm$, and the spin-only group is $\{C_{\infty z'}, M_x' C_{\infty z'}\}$, indicating a collinear magnetic configuration.

POSCAR.symm

POSCAR.symm is the result of symmetrizing the input POSCAR according to SSG symmetry operations, which is closer to the identified SSG symmetry than the input POSCAR.

ssg.data

ssg.data is a binary file storing SSG operations and other information, used by the `-pw` mode or `-wann` mode.

3.2 irssg -pw: SSG Band Coirrep Analysis Based on Plane Waves (VASP)

After obtaining VASP results without spin-orbit coupling (OUTCAR and WAVECAR), the `-pw` mode can be used to directly perform coirrep analysis on the bands.

3.2.1 Input Files

The input files for the `-pw` mode are VASP output files `OUTCAR` and `WAVECAR`, as well as `ssg.data` obtained from the `-ssg` mode.

For VASP calculations with noncollinear magnetic moments, `INCAR` should set `LNONCOLLINEAR=.TRUE.` and `LSORBIT=.FALSE.`. For collinear magnetic moments, the same settings can also be used, but in that case the up-spin and down-spin bands cannot be separated, so the output format of the `-pw` mode is identical to that for noncollinear cases.

Alternatively, for collinear cases, `INCAR` can use `LNONCOLLINEAR=.FALSE.` and `ISPIN=2`, which allows separation of up-spin and down-spin bands. If there are no SSG little-group operations that flip spins at a k point, the `-pw` mode computes representations separately for up and down spins. If there are SSG little-group operations that flip spins at a k point, they are computed together.

3.2.2 Command

```
$ irssg -pw [PW-related options]
```

Before entering the `-pw` mode, if `ssg.data` is missing in the current directory, `irssg` will automatically call the `-ssg` mode once (i.e., execute `irssg -ssg`, optionally with SSG-related parameters you provide on the command line).

Common options supported by the plane-wave mode:

- `-nk k_start k_end`: Process only k points from `k_start` to `k_end` (numbered from 1); if omitted, all k points are processed by default;
- `-nb band_min band_max`: Restrict the band range for coirrep analysis:
 - `band_min`: lowest band index;
 - `band_max`: highest band index.

If not specified, all DFT bands in the current calculation are included by default;

- `-tolE tol`: Energy tolerance (corresponding to `tolE` in the code, default 1.0×10^{-3}), used for judging degeneracies and constructing coirreps:
 - To distinguish energy levels more strictly, reduce `tol`;
 - If there is numerical noise in the energies, increase `tol` appropriately.

Examples:

```
$ irssg -pw > irssg.out
$ irssg -pw -nk 5 20 -nb 10 30 -tolE 1.0e-4 > irssg.out
```

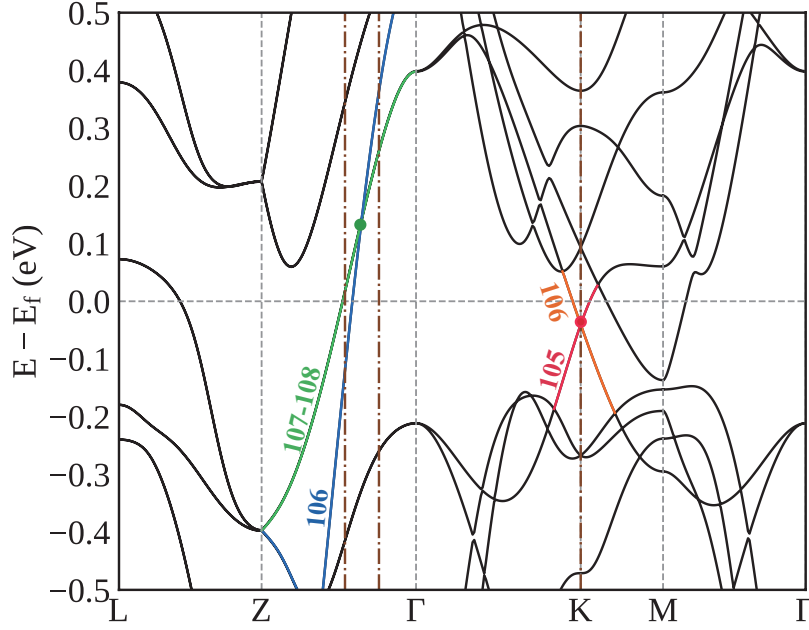


Figure 1: Magnetic electronic band structure of Mn_3Sn and its representations.

3.2.3 Output Files

irssg.out (screen output)

The `irssg.out` file will sequentially output: all operations of $\mathcal{G}/\mathcal{T}_0$, fractional coordinates and names of specified k points, unitary and anti-unitary parts of the little group at specified k points, character tables and phases of irreducible projective representations of the unitary subgroup of the little group at specified k points, irreducible projective coirreps of the little group at specified k points, characters of bands under SSG operations, and representation symbols of the bands.

A typical example for Mn_3Sn :

```
It belongs to Spin Space Group (SSG)  $P\{1,1,1\} 6_3\{3^{-1}\} / m\bar{1} m\bar{m} c\bar{m}$  (C3v-II) .
the list of 48 SSG operators: { Ui || Ri | tau_i }
Ui(03, spin part)   Ri   tau_i   Ri(Cart. Coord.)   SU2(spin part)
# 1 without time reversal
1.000 0.000 0.000   1 0 0   0.000   1.000 0.000 0.000   ( 1.000 0.000 ) ( 0.000 0.000 )
0.000 1.000 0.000   0 1 0   0.000   0.000 1.000 0.000   ( 0.000 0.000 ) ( 1.000 0.000 )
0.000 0.000 1.000   0 0 1   0.000   0.000 0.000 1.000
.....
# 48 without time reversal
-0.500 0.866 0.000  -1 1 0  -0.000  -1.000 0.000 0.000   ( 0.000-0.000 ) (-0.866-0.500)
 0.866 0.500 0.000   0 1 0   0.000   0.000 1.000 0.000   ( 0.866-0.500 ) ( 0.000 0.000 )
 0.000 0.000-1.000   0 0 1   0.000   0.000 0.000 1.000
*****
knum = 1
K = 0.000000 0.000000 0.000000
kname = GM   Fractional coordinate: 0.00 0.00 0.00 (given in the conventional basis)

24 unitary elements of the little group:
 1  3  5  7  9 11 13 15 17 19 21 23 26 28 30
32 34 36 38 40 42 44 46 48
24 anti-unitary elements of the little group:
```

```

  2   4   6   8  10  12  14  16  18  20  22  24  25  27  29
31  33  35  37  39  41  43  45  47

Character table for unitary group
tor      1      3      5      .....      44      48
GM1  1  1.00+0.00i  1.00+0.00i  1.00-0.00i  .....  0.00-1.00i  0.00-1.00i
.....
GM12  1  2.00+0.00i -2.00+0.00i -1.00-0.00i  .....  0.00-0.00i  0.00+0.00i
Coirreps for complete group: GM1,GM2,GM3,GM4,GM5,GM6,GM7,GM8,GM9,GM10,GM11,GM12

Band Characters and Co-representations:
bnd ndg  eigval      1      3      ... ..      46      48
  1  1 -0.4054E+02  1.00+0.00i -1.00+0.00i  ... ..  0.00+1.00i  0.00-1.00i  = GM7
.....
108  1  0.9716E+01  1.00+0.00i -1.00-0.00i  ... ..  0.00+1.00i  0.00-1.00i  = GM7
*****
knum = 2
K = 0.000000 0.000000 0.230000
kname = DT Fractional coordinate: 0.00 0.00 w (given in the conventional basis)

12 unitary elements of the little group:
  1   5   9  13  17  21  28  32  36  40  44  48
12 anti-unitary elements of the little group:
  4   8  12  16  20  24  25  29  33  37  41  45

Character table for unitary group
tor      1      5      9      .....      44      48
DT1  1  1.00+0.00i  1.00+0.00i -1.00+0.00i  .....  0.00-1.00i  0.00-1.00i
.....
DT6  1  2.00+0.00i -1.00-0.00i  1.00-0.00i  .....  0.00+0.00i  0.00-0.00i
phase 1.00+0.00i  0.75+0.66i  1.00+0.00i  .....  0.75+0.66i  1.00+0.00i
Coirreps for complete group: DT1,DT2,DT3,DT4,DT5,DT6

Band Characters and Co-representations:
bnd ndg  eigval      1      5      ... ..      44      48
  1  3 -0.4053E+02  3.00+0.00i -1.50-1.32i  ... .. -0.66+0.75i  0.00-1.00i  = DT4 + DT6
.....
104  2  0.8713E+01  2.00+0.00i  0.75+0.66i  ... ..  0.00+0.00i  0.00-0.00i  = DT5
106  1  0.9008E+01  1.00+0.00i -0.75-0.66i  ... ..  0.00-1.00i -0.66+0.75i  = DT4
107  2  0.9145E+01  2.00+0.00i -0.75-0.66i  ... ..  0.00+0.00i  0.00-0.00i  = DT6
*****
knum = 3
K = 0.000000 0.000000 0.120000
kname = DT Fractional coordinate: 0.00 0.00 w (given in the conventional basis)

12 unitary elements of the little group:
  1   5   9  13  17  21  28  32  36  40  44  48
12 anti-unitary elements of the little group:
  4   8  12  16  20  24  25  29  33  37  41  45

Character table for unitary group
tor      1      5      9      .....      44      48
DT1  1  1.00+0.00i  1.00+0.00i -1.00+0.00i  .....  0.00-1.00i  0.00-1.00i
.....
DT6  1  2.00+0.00i -1.00-0.00i  1.00-0.00i  .....  0.00+0.00i  0.00-0.00i
phase 1.00+0.00i  0.75+0.66i  1.00+0.00i  .....  0.75+0.66i  1.00+0.00i
Coirreps for complete group: DT1,DT2,DT3,DT4,DT5,DT6

Band Characters and Co-representations:
bnd ndg  eigval      1      5      ... ..      44      48
  1  1 -0.4053E+02  1.00+0.00i -0.93-0.37i  ... .. -0.37+0.93i  0.00-1.00i  = DT4
.....
104  2  0.8868E+01  2.00+0.00i  0.93+0.37i  ... ..  0.00+0.00i  0.00-0.00i  = DT5
106  2  0.9389E+01  2.00+0.00i -0.93-0.37i  ... ..  0.00-0.00i  0.00+0.00i  = DT6
108  1  0.9492E+01  1.00+0.00i -0.93-0.37i  ... ..  0.00-1.00i -0.37+0.93i  = DT4
*****
knum = 4
K = 0.333333 0.333333 0.000000
kname = K Fractional coordinate: 0.33 0.33 0.00 (given in the conventional basis)

12 unitary elements of the little group:
  1   7   9  15  17  23  28  30  36  38  44  46
12 anti-unitary elements of the little group:
  4   6  12  14  20  22  25  31  33  39  41  47

Character table for unitary group
tor      1      7      9      .....      44      46
K1  1  1.00+0.00i  1.00-0.00i -1.00-0.00i  .....  0.00-1.00i  0.00-1.00i
.....
K6  1  2.00+0.00i -1.00+0.00i  1.00+0.00i  .....  0.00-0.00i  0.00+0.00i

```

```

Coirreps for complete group: K1,K2,K3,K4,K5,K6

Band Characters and Co-representations:
bnd ndg   eigval      1      7      ...      44      46
  1  1 -0.4054E+02    1.00+0.00i  1.00-0.00i  ...  0.00+1.00i  0.00+1.00i  = K2
.....
103  2  0.8861E+01    2.00+0.00i  1.00-0.00i  ...  0.00+0.00i  0.00-0.00i  = K5
105  2  0.9088E+01    2.00+0.00i -1.00+0.00i  ...  0.00+0.00i  0.00-0.00i  = K6
107  2  0.9221E+01    2.00+0.00i  1.00+0.00i  ...  0.00+0.00i  0.00-0.00i  = K5
*****
TOTAL END

```

Plotting the bands and band representations obtained for Mn_3Sn , as shown in Fig. 1, one can see an apparent crossing along Z Γ between band 106 and bands 107/108. According to the obtained representations, this is a three-fold crossing formed by the one-dimensional representation DT4 and the two-dimensional representation DT6. In addition, at the K point, bands 105 and 106 appear to cross; the obtained representation shows that this point is a two-dimensional coirrep, indicating an ordinary symmetry-protected degeneracy.

For $\text{KV}_2\text{Se}_2\text{O}$, we set the INCAR parameters LNONCOLLINEAR=.FALSE. and ISPIN=2 for calculation. An example output is as follows:

```

It belongs to Spin Space Group (SSG)  $P\{1,1,1\} 4\bar{m} / m\bar{1} m\bar{1} m\bar{1} (\text{Cs}\bar{I})$  .
the list of 64 SSG operators: { Ui || Ri | tau_i }
  Ui(03, spin part)      Ri      tau_i      Ri(Cart. Coord.)      SU2(spin part)
#  1  without time reversal
  1.000 0.000 0.000  1 0 0  0.000  1.000 0.000 0.000  ( 1.000 0.000) ( 0.000 0.000)
  0.000 1.000 0.000  0 1 0  0.000  0.000 1.000 0.000  ( 0.000 0.000) ( 1.000 0.000)
  0.000 0.000 1.000  0 0 1  0.000  0.000 0.000 1.000
.....
# 64  without time reversal
  1.000 0.000 0.000  0-1 0  0.000  0.000-1.000 0.000  ( 0.000 0.000) ( 0.000-1.000)
  0.000-1.000 0.000  -1 0 0  0.000  -1.000 0.000 0.000  ( 0.000-1.000) ( 0.000 0.000)
  0.000 0.000-1.000  0 0 1  0.000  0.000 0.000 1.000
*****
There is a SSG operator flipping spin in the k-little group.
knum =  1 spin = up \& down
K = 0.000000 0.000000 0.000000
kname = GM      Fractional coordinate: 0.00 0.00 0.00 (given in the conventional basis)

32 unitary elements of the little group:
  1  2  5  6  11  12  15  16  17  18  21  22  27  28  31
.....
32 anti-unitary elements of the little group:
  3  4  7  8  9  10  13  14  19  20  23  24  25  26  29
.....

Character table for unitary group
      tor      1      2      .....      63      64
GM1  1  2.00+0.00i  0.00+0.00i  .....  0.00+0.00i  0.00+0.00i
.....
GM8  1  2.00+0.00i  0.00-0.00i  .....  0.00-0.00i  0.00+0.00i
Coirreps for complete group: GM1,GM2,GM3,GM4,GM5,GM6,GM7,GM8

Band Characters and Co-representations:
bnd ndg   eigval      1      2      ...      63      64
  1  2 -0.6011E+02    2.00+0.00i  0.00-0.00i  ...  0.00-0.00i  0.00-0.00i  = GM1
.....
127  2  0.2323E+02    2.00+0.00i  0.00-0.00i  ...  0.00-0.00i  0.00-0.00i  = GM1
*****
knum =  2 spin = up
K = 0.000000 0.500000 0.000000
kname = X      Fractional coordinate: 0.00 0.50 0.00 (given in the conventional basis)

16 unitary elements of the little group:
  1  2  5  6  17  18  21  22  33  34  37  38  49  50  53
  54
16 anti-unitary elements of the little group:
  3  4  7  8  19  20  23  24  35  36  39  40  51  52  55
  56

```

```

Character table for unitary group
  tor      1      2      .....      53      54
X1  1      1.00+0.00i  0.00+1.00i  .....  1.00-0.00i  0.00+1.00i
.....
X16 1      1.00+0.00i  0.00-1.00i  ..... -1.00+0.00i  0.00+1.00i
Coirreps for complete group: X1,X2,X3,X4,X5,X6,X7,X8,X9,X10,X11,X12,X13,X14,X15,X16

Band Characters and Co-representations:
  bnd ndg   eigval      1      2      ... ..      53      54
  1  1 -0.6011E+02    1.00+0.00i  0.00-1.00i  ... ..  1.00-0.00i  0.00-1.00i  = X9
.....
  54 1  0.1876E+02    1.00+0.00i  0.00-1.00i  ... .. -1.00+0.00i  0.00+1.00i  = X13
*****
There is a SSG operator flipping spin in the k-little group.
knum = 1 spin = up \& down
K = 0.000000 0.000000 0.000000
kname = GM Fractional coordinate: 0.00 0.00 0.00 (given in the conventional basis)

32 unitary elements of the little group:
  1  2  5  6 11 12 15 16 17 18 21 22 27 28 31
.....
32 anti-unitary elements of the little group:
  3  4  7  8  9 10 13 14 19 20 23 24 25 26 29
.....

Character table for unitary group
  tor      1      2      .....      63      64
GM1 1      2.00+0.00i  0.00-0.00i  .....  0.00+0.00i  0.00-0.00i
.....
GM8 1      2.00+0.00i  0.00-0.00i  .....  0.00+0.00i  0.00-0.00i
Coirreps for complete group: GM1,GM2,GM3,GM4,GM5,GM6,GM7,GM8

Band Characters and Co-representations:
  bnd ndg   eigval      1      2      ... ..      63      64
  1  2 -0.6011E+02    2.00+0.00i  0.00-0.00i  ... ..  0.00-0.00i  0.00-0.00i  = GM1
.....
 127 2  0.2323E+02    2.00+0.00i  0.00-0.00i  ... ..  0.00-0.00i  0.00-0.00i  = GM1
*****
knum = 2 spin = down
K = 0.000000 0.500000 0.000000
kname = X Fractional coordinate: 0.00 0.50 0.00 (given in the conventional basis)

16 unitary elements of the little group:
  1  2  5  6 17 18 21 22 33 34 37 38 49 50 53
  54
16 anti-unitary elements of the little group:
  3  4  7  8 19 20 23 24 35 36 39 40 51 52 55
  56

Character table for unitary group
  tor      1      2      .....      53      54
X1  1      1.00+0.00i  0.00+1.00i  .....  1.00+0.00i  0.00+1.00i
.....
X16 1      1.00+0.00i  0.00-1.00i  ..... -1.00-0.00i  0.00+1.00i
Coirreps for complete group: X1,X2,X3,X4,X5,X6,X7,X8,X9,X10,X11,X12,X13,X14,X15,X16

Band Characters and Co-representations:
  bnd ndg   eigval      1      2      ... ..      53      54
  1  1 -0.6011E+02    1.00+0.00i  0.00-1.00i  ... .. -1.00-0.00i  0.00+1.00i  = X16
.....
  64 1  0.2590E+02    1.00+0.00i  0.00-1.00i  ... ..  1.00+0.00i  0.00-1.00i  = X12
*****
TOTAL END

```

It can be seen that in the `irssg.out` file, the band coirreps for up-spin bands are output first, followed by those for down-spin bands, with each k point output twice. If there are little-group operations that can flip spins, the message ‘There is a SSG operator flipping spin in the k-little group.’ will be output before this entire block, and the spin in both outputs will be displayed as ‘spin = up & down’, for example at Γ ; if no such operations exist, the spin in the two outputs will be displayed separately as ‘spin = up’ and ‘spin = down’ with corresponding band coirreps computed, for example at X.

`chart.dat`

The `chart.dat` file lists the names of each high-symmetry point/line/plane, spin space group little-group elements, character tables of irreducible projective representations of the unitary subgroup of the little group, and compatibility relations between little groups of adjacent k points.

A typical example for Mn_3Sn :

```
kname = GM      Fractional coordinate: 0.00 0.00 0.00 (given in the conventional basis)

24 unitary elements of the little group:
  1   3   5   7   9  11  13  15  17  19  21  23  26  28  30
32  34  36  38  40  42  44  46  48
24 anti-unitary elements of the little group:
  2   4   6   8  10  12  14  16  18  20  22  24  25  27  29
31  33  35  37  39  41  43  45  47

Character table for unitary group
      tor      1      3      5      .....      46      48
GM1   1      1.00+0.00i  1.00+0.00i  1.00-0.00i  .....  0.00-1.00i  0.00-1.00i
.....
GM12  1      2.00+0.00i -2.00+0.00i -1.00-0.00i  .....  0.00-0.00i  0.00+0.00i
Coirreps for complete group: GM1,GM2,GM3,GM4,GM5,GM6,GM7,GM8,GM9,GM10,GM11,GM12

kname = DT      Fractional coordinate: 0.00 0.00 w (given in the conventional basis)

12 unitary elements of the little group:
  1   5   9  13  17  21  28  32  36  40  44  48
12 anti-unitary elements of the little group:
  4   8  12  16  20  24  25  29  33  37  41  45

Character table for unitary group
      tor      1      5      9      .....      44      48
DT1   1      1.00+0.00i  1.00+0.00i -1.00+0.00i  .....  0.00-1.00i  0.00-1.00i
.....
DT6   1      2.00+0.00i -1.00-0.00i  1.00-0.00i  .....  0.00+0.00i  0.00-0.00i
Coirreps for complete group: DT1,DT2,DT3,DT4,DT5,DT6

kname = K       Fractional coordinate: 0.33 0.33 0.00 (given in the conventional basis)

12 unitary elements of the little group:
  1   7   9  15  17  23  28  30  36  38  44  46
12 anti-unitary elements of the little group:
  4   6  12  14  20  22  25  31  33  39  41  47

Character table for unitary group
      tor      1      7      9      .....      44      46
K1    1      1.00+0.00i  1.00-0.00i -1.00-0.00i  .....  0.00-1.00i  0.00-1.00i
.....
K6    1      2.00+0.00i -1.00+0.00i  1.00+0.00i  .....  0.00-0.00i  0.00+0.00i
Coirreps for complete group: K1,K2,K3,K4,K5,K6

GM->DT Compatibility relations:
GM1->DT1
GM2->DT2
.....
GM12->DT6
```

fort.180

This file records the characters of band representations at each k point under all SSG operations.

3.3 irssg -wann: SSG Band Coirrep Analysis Based on Tight-Binding Hamiltonians (Wannier)

If a Wannier tight-binding Hamiltonian (for example `case_hr.dat`) has been constructed and the corresponding `tbbox.in` is prepared, the `-wann` mode can be used.

3.3.1 Input Files

The `-wann` mode has two input files: `case_hr.dat` and `tbbox.in`. Here, `case_hr.dat` is a Hamiltonian file generated by Wannier90, or a self-constructed Hamiltonian file in the same format as `wannier90_hr.dat`; the name can be arbitrary but must be consistent with the parameters in `tbbox.in`. `tbbox.in` is the master input file for the `-wann` mode that describes the tight-binding model and orbital projection information. A typical form is as follows (Mn_3Sn):

```
spinpol = False
#hr_name_up = sp.up_hr.dat
#hr_name_dn = sp.dn_hr.dat
hr_name = symm_hr.dat

proj:
orbt = 2
spincov = 1
ntau = 8
  0.838800  0.677599  0.250000  1 5 ! x1,x2,x3,itauiorbit
  0.161200  0.838800  0.750000  1 5
  0.838800  0.161200  0.250000  1 5
  0.161200  0.322401  0.750000  1 5
  0.322401  0.161200  0.250000  1 5
  0.677599  0.838800  0.750000  1 5
  0.333333  0.666667  0.250000  2 3
  0.666667  0.333333  0.750000  2 3
end projections

kpoint:
kmesh = 10
Nk = 2
  0 0 0
  0.5 0 0
end kpoint

unit_cell:
  5.665000  0.000000  0.000000
 -2.832500  4.906034  0.000000
  0.000000  0.000000  4.531000
end unit
```

The meanings of the main blocks and key parameters are as follows:

- `spinpol`: Whether to use a spin-polarized tight-binding model; `False` indicates a single Hamiltonian (spin-independent), while `True` requires separate provision of `hr_name_up` / `hr_name_dn`.
- `hr_name` / `hr_name_up` / `hr_name_dn`: Names of `*_hr.dat` files generated by Wannier90, in the same format as `wannier90_hr.dat`.
- `unit_cell ... end unit`: 3×3 real-space lattice basis vector matrix (units in Å), where each row is one primitive lattice vector, which can be directly copied from lines 3–5 of POSCAR.

- `kpoint ... end kpoint`: Defines the k -space path for band plotting:
 - `kmesh`: Number of evenly spaced samples between each pair of consecutive listed k -nodes;
 - `Nk`: Number of k points that follow; the path is formed by connecting them in order;
 - Each line is a k point given in fractional coordinates with respect to the reciprocal basis vectors ($\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$).
- `proj ... end projections`: Defines how local orbitals are projected onto Wannier orbitals:
 - `orbt`: Controls the ordering convention for local orbitals; different values correspond to different (s, p, d, f) combination orders (compatible with Wannier90 conventions); for more detailed correspondence between `iorbit` and specific orbitals, see Table 1.
 - `spincov`: Ordering of the spin-orbital tensor product; 1 means “orbit-major, spin-minor” (list all up-spin orbitals first, then all down-spin orbitals), 2 means “spin-major, orbit-minor” (for each orbital, list up-spin first, then down-spin);
 - `ntau`: Number of TB orbital (position) sites listed in this block;
 - Each line `x1 x2 x3 itau iorbit`: `x1,x2,x3` are fractional coordinates of atoms in the primitive cell, `itau` labels the atomic species or Wyckoff position number, and `iorbit` selects the orbital combination type on this atom (determined by `orbt` and a preset table, for example pure s , p triplet, d quintet, or their combinations).

Additionally, consider Wannier Hamiltonians calculated with INCAR parameters `LNONCOLLINEAR=.FALSE.` and `ISPIN=2`. Using $\text{KV}_2\text{Se}_2\text{O}$ as an example, the `tbbox.in` file is as follows:

```
spinpol = True
hr_name_up = sp.up_hr.dat
hr_name_dn = sp.dn_hr.dat

proj:
orbt = 2
spincov = 1
ntau = 6
  0.000000  0.000000  0.500000  1  1
  0.500000  0.000000  0.000000  2  5
  0.000000  0.500000  0.000000  2  5
  0.500000  0.500000  0.240450  3  3
  0.500000  0.500000  0.759550  3  3
  0.000000  0.000000  0.000000  4  3
end projections

kpoint:
kmesh = 1
Nk = 2
  0 0 0
  0.500000  0.000000  0.000000 ! k0: y1,y2,y3
end kpoint

unit_cell:
  3.951940  0.000000  0.000000
  0.000000  3.951940  0.000000
  0.000000  0.000000  7.352430
end unit
```

Since calculations with INCAR parameters `LNONCOLLINEAR=.FALSE.` and `ISPIN=2` produce separate Wannier Hamiltonians for up-spin and down-spin, the file names must be specified separately in `hr_name_up` and `hr_name_dn`.

Table 1: Local orbital ordering in different conventions.

iorbit	Convention 1	Convention 2 (Wannier90 convention)
1	s	s
3	p_x, p_y, p_z	p_z, p_x, p_y
5	$d_{xy}, d_{yz}, d_{xz}, d_{x^2-y^2}, d_{z^2}$	$d_{z^2}, d_{xz}, d_{yz}, d_{x^2-y^2}, d_{xy}$
4	s, p_x, p_y, p_z	s, p_z, p_x, p_y
6	$s, d_{xy}, d_{yz}, d_{xz}, d_{x^2-y^2}, d_{z^2}$	$s, d_{z^2}, d_{xz}, d_{yz}, d_{x^2-y^2}, d_{xy}$
8	$p_x, p_y, p_z, d_{xy}, d_{yz}, d_{xz}, d_{x^2-y^2}, d_{z^2}$	$p_z, p_x, p_y, d_{z^2}, d_{xz}, d_{yz}, d_{x^2-y^2}, d_{xy}$
9	$s, p_x, p_y, p_z, d_{xy}, d_{yz}, d_{xz}, d_{x^2-y^2}, d_{z^2}$	$s, p_z, p_x, p_y, d_{z^2}, d_{xz}, d_{yz}, d_{x^2-y^2}, d_{xy}$
7	$f_{xyz}, f_{5x^3-3xr^2}, f_{5y^3-3yr^2}, f_{5z^3-3zr^2}, f_{x(y^2-z^2)}, f_{y(z^2-x^2)}, f_{z(x^2-y^2)}$	$f_{z^3}, f_{xz^2}, f_{yz^2}, f_{z(x^2-y^2)}, f_{xyz}, f_{x(x^2-3y^2)}, f_{y(3x^2-y^2)}$

3.3.2 Command

```
$ irssg -wann [Wannier-related options]
```

In the `-wann` mode, if `ssg.data` is missing in the current directory, SSG analysis will be automatically called first (equivalent to `irssg -ssg`).

The `-wann` mode shares the same set of band-selection parameters with the `-pw` mode:

- `-nk k_start k_end`: Process only a subset of k points;
- `-nb band_min band_max`: Analyze only a specified band window;
- `-tolE tol`: Energy tolerance.

Examples:

```
$ irssg -wann > irssg.out
$ irssg -wann -nb 10 30 > irssg.out
```

3.3.3 Output Files

The `irssg.out`, `chart.dat`, and `fort.180` files generated by the `-wann` mode have identical content and format to those of the `-pw` mode in the previous subsection. For specific field meanings and examples, please refer to Sec. 3.2.3.

3.4 Default: No Mode Options

When running `irssg` directly in the command line without any mode options (`-ssg` / `-pw` / `-wann`), the default behavior of the program is:

- First, it is equivalent to executing `irssg -ssg` once and generating SSG information together with `ssg.data`;
- If both `OUTCAR` and `WAVECAR` exist in the current directory, after completing SSG analysis, automatically enter plane-wave mode (equivalent to continuing with `irssg -pw`, with parameter meanings as described above).